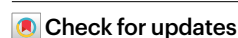


Quantum recoil in free-electron interactions with atomic lattices

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The emission of light from charged particles underlies a wealth of scientific phenomena and technological applications. Classical theory determines the emitted photon energy by assuming an undeflected charged particle trajectory. In 1940, Ginzburg pointed out that this assumption breaks down in quantum electrodynamics, resulting in shifts—known as quantum recoil—in outgoing photon energies from their classically predicted values. Since then, quantum recoil in free-electron light-emission processes, including Cherenkov radiation and Smith–Purcell radiation, has been well-studied in theory, but an experimental demonstration has remained elusive. Here we present an experimental demonstration of quantum recoil, showing that this quantum electrodynamical effect is not only observable at room temperature but also robust in the presence of other electron-scattering mechanisms. By scattering free electrons off the periodic two-dimensional atomic sheets of van der Waals materials in a tabletop platform, we show that the X-ray photon energy is accurately predicted only by quantum recoil theory. We show that quantum recoil can be enormous, to the point that a classically predicted X-ray photon is emitted as an extremely low-energy photon. We envisage quantum recoil as a means of precision control over outgoing photon and electron spectra, and show that quantum recoil can be tailored through a host of parameters: the electron energy, the atomic composition and the tilt angle of the van der Waals material. Our results pave the way to tabletop, room-temperature platforms for harnessing and investigating quantum electrodynamical effects in electron–photon interactions.

Free-electron spontaneous emission phenomena, such as the Smith–Purcell effect^{1,2} and Cherenkov radiation^{3,4}, have driven a wide range of technological innovations and scientific discoveries^{5–19}. In the Smith–Purcell effect, photons are emitted when the Coulomb field of an electron scatters off a periodic structure as the electron travels in constant motion near to or through it. The Smith–Purcell effect has been investigated as a tunable light source in many regimes of the electromagnetic spectrum, from microwave to X-ray frequencies^{19–22}.

In addition, the Smith–Purcell effect has been leveraged for particle identification^{13,14}, particle acceleration²³ and the discovery of fundamental radiation limits^{10,24}.

Classical theory predicts that the deceleration of the incident electron upon photon emission has a negligible effect in Smith–Purcell radiation. As such, the emitted photon energies predicted by classical theory are a first-order approximation, which does not take into consideration energy–momentum conservation enforced by quantum

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electrodynamics. As pointed out theoretically by Ginzburg in 1940²⁵, energy–momentum conservation results in quantum recoil, namely, shifts in the emitted photon energies from the classically predicted values. Furthermore, the emitted photon energies are no longer evenly spaced (that is, they are no longer integer multiples of some fundamental harmonic), as is the case in classical theory.

Here we experimentally demonstrate quantum recoil. In particular, we theoretically predict and experimentally measure quantum recoil in Smith–Purcell radiation using electrons with kinetic energies of 10–15 keV from a scanning electron microscope (SEM). The platform is therefore laboratory-scale, with no external light sources involved. The electrons scatter off the subnanometre periodic gratings provided by the multilayer two-dimensional atomic sheets in van der Waals materials such as graphite and hexagonal boron nitride. We are able to observe measurable photon energy shifts in the emitted photons. These experimental measurements confirm the accuracy and importance of quantum recoil theory. We show that quantum recoil can be tailored by controlling the electron energy, the target material's atomic composition and tilt angle, and the emission order. We show that the versatility of quantum recoil enables a classically predicted soft X-ray photon to be shown as a low-energy photon through precise control of the electron and material parameters. We conclude that quantum recoil can be substantial, even when driven using conventional tabletop electron sources at room temperature, as well as in the presence of other electron-scattering events. Even more broadly, our results suggest that the combination of existing tabletop electron sources and crystalline materials is a promising platform for studying related fundamental phenomena, including open questions such as radiation reaction²⁶ and intriguing concepts like the quantum free-electron laser^{27,28}. Our results also highlight the importance of taking quantum recoil into account when modelling free-electron-driven material systems such as van der Waals-based X-ray emitters^{21,22} and other coherent X-ray sources^{29–32}.

Results

Free electrons impinging on a crystalline solid (Fig. 1a) emit radiation via two mechanisms: the scattering of the free-electron Coulomb fields off the periodic lattice, which is known as parametric X-ray radiation^{31,32}; and the acceleration of the free electrons in the crystal's electrostatic potential, which is known as coherent bremsstrahlung^{30–33}. Parametric X-ray radiation, the dominant mechanism in our regime of study, is physically equivalent to the Smith–Purcell effect², with the atomic lattice planes playing the role of the periodic grating. In the classical picture where quantum recoil is ignored, the momentum of the electron is assumed to be unchanged upon photon emission, and the output photon angular frequency here, ω_c , is given by (for example, ref. 30)

$$\omega_c = -\frac{c\mathbf{g} \cdot \boldsymbol{\beta}}{1 - n\boldsymbol{\beta} \cdot \hat{\mathbf{q}}}, \quad (1)$$

where c is the speed of light in free space, $\mathbf{g}$ is the reciprocal lattice vector (whose magnitude is g), $\boldsymbol{\beta} \equiv \mathbf{v}/c$ is the normalized electron velocity (where $\mathbf{v}$ is the electron velocity), n is the refractive index of the medium and $\hat{\mathbf{q}} = (\sin\theta_{\text{obs}}\cos\phi_{\text{obs}}, \sin\theta_{\text{obs}}\sin\phi_{\text{obs}}, \cos\theta_{\text{obs}})$ is the unit vector of the emitted photon wave vector (where θ_{obs} and ϕ_{obs} are the polar and azimuthal angle of $\hat{\mathbf{q}}$, respectively). In many cases, even up to recent times, equation (1) has been shown to be adequate for explaining experimental observations^{20,22,34,35}.

As early as 1940, however, Vitaly L. Ginzburg pointed out theoretically that the above consideration alone cannot describe the free-electron spontaneous emission with full accuracy²⁵. This is because the classical picture ignores quantum recoil, which refers specifically to the $\hbar$ -dependent photon energy shift that results from the electron's reaction upon spontaneously emitting the photon in question (Fig. 1b).

Quantum recoil is therefore completely different from—and highly complementary to—other types of recoil, such as those associated with the deflection of electrons by external fields^{7,36,37} and the deflection of electrons by scattering events before and after the spontaneous emission event in question^{36,38}. Ginzburg's theory has since been expanded by others in the context of Cherenkov radiation and Smith–Purcell radiation^{25,39–42}. By applying energy–momentum conservation to our Smith–Purcell scenario, we obtain the output photon angular frequency, ω , as

$$\omega = -\frac{2c\mathbf{g} \cdot \boldsymbol{\beta} + \hbar c^2 g^2/E_i}{(1 - n\boldsymbol{\beta} \cdot \hat{\mathbf{q}} - n\hbar c\mathbf{g} \cdot \hat{\mathbf{q}}/E_i) \left(1 + \sqrt{1 - \frac{n^2 - 1}{n^2} \frac{\hbar^2 c^2 g^2 + 2\hbar c E_i \mathbf{g} \cdot \boldsymbol{\beta}}{[E_i(\boldsymbol{\beta} \cdot \hat{\mathbf{q}} - 1/n) + \hbar c\mathbf{g} \cdot \hat{\mathbf{q}}]^2}} \right)}, \quad (2)$$

where E_i is the total energy (that is, including the rest mass) of the incident electron. Equation (2) reduces to the classical equation (1) when the reduced Planck's constant $\hbar$ is artificially set to zero. The parts containing $\hbar$ thus constitute deviations from the classical prediction of equation (1). Indeed, the quantum nature of quantum recoil can be seen from its dependence on $\hbar$, which is widely regarded as the signature of quantum mechanics. We note that the appearance of $\hbar$ in equation (2) is not due to quantization of the photon energy and momentum (that is, the de Broglie relations) but is instead due to energy–momentum conservation in a quantum electrodynamical process. We point out that a similar result to equation (2) was obtained in ref. 39, albeit for the specific cases of one- and two-dimensional nanogratings. It can be shown that equation (2) agrees with the result in ref. 39 under those specialized conditions. The refractive index in the X-ray range can be modelled as $n = 1 - \frac{\omega_p^2}{2\omega^2} \approx 1$, where ω_p is the plasma frequency, using the Lorentz oscillator model. We therefore simplify equation (2) to

$$\omega \approx -\frac{c\mathbf{g} \cdot \boldsymbol{\beta} + \hbar c^2 g^2/(2E_i)}{1 - \boldsymbol{\beta} \cdot \hat{\mathbf{q}} - \hbar c\mathbf{g} \cdot \hat{\mathbf{q}}/E_i}. \quad (3)$$

In the limit where quantum recoil is small, we can Taylor expand equation (3) to obtain

$$\omega \approx -\frac{c\mathbf{g} \cdot \boldsymbol{\beta}}{1 - \boldsymbol{\beta} \cdot \hat{\mathbf{q}}} + \hbar \frac{c^2 \mathbf{g} \cdot \hat{\mathbf{q}}(\mathbf{g} \cdot \boldsymbol{\beta})/E_i + c^2 g^2(1 - \boldsymbol{\beta} \cdot \hat{\mathbf{q}})/(2E_i)}{(1 - \boldsymbol{\beta} \cdot \hat{\mathbf{q}})^2}, \quad (4)$$

where the presence of $\hbar$ in the second term indicates the quantum nature of the quantum recoil correction. In many cases, however, quantum recoil is substantial and equation (3) cannot be simplified in this way. We show below that there are conditions under which quantum recoil can be so large that a classically predicted X-ray photon is in fact emitted as a low-energy photon. Details of the derivations are given in Supplementary Section 1. The experiments are performed using a SEM setup, where electrons with a kinetic energy 10–15 keV impinge on single-crystal van der Waals materials placed in the sample holder (see Methods). The van der Waals material (graphite in the case of Fig. 1) is oriented such that the electron travels roughly parallel to the [001] direction. The grating vector $\mathbf{g}$ is thus given by

$$\mathbf{g}_{(00m)} = -m \frac{2\pi}{d} (\sin\theta_{\text{til}} \cos\phi_{\text{til}}, \sin\theta_{\text{til}} \sin\phi_{\text{til}}, \cos\theta_{\text{til}}), \quad (5)$$

where m is an integer denoting the order of the Smith–Purcell radiation, d is the interlayer distance of the (001) planes, and θ_{til} and ϕ_{til} , respectively, denote the polar and azimuthal angles of the [001] zone axis relative to the initial direction of electron travel (see Fig. 1c inset). For hexagonal crystals such as graphite and hexagonal boron nitride, emission at odd m is forbidden since the structure factor is zero (see Supplementary Section 2).

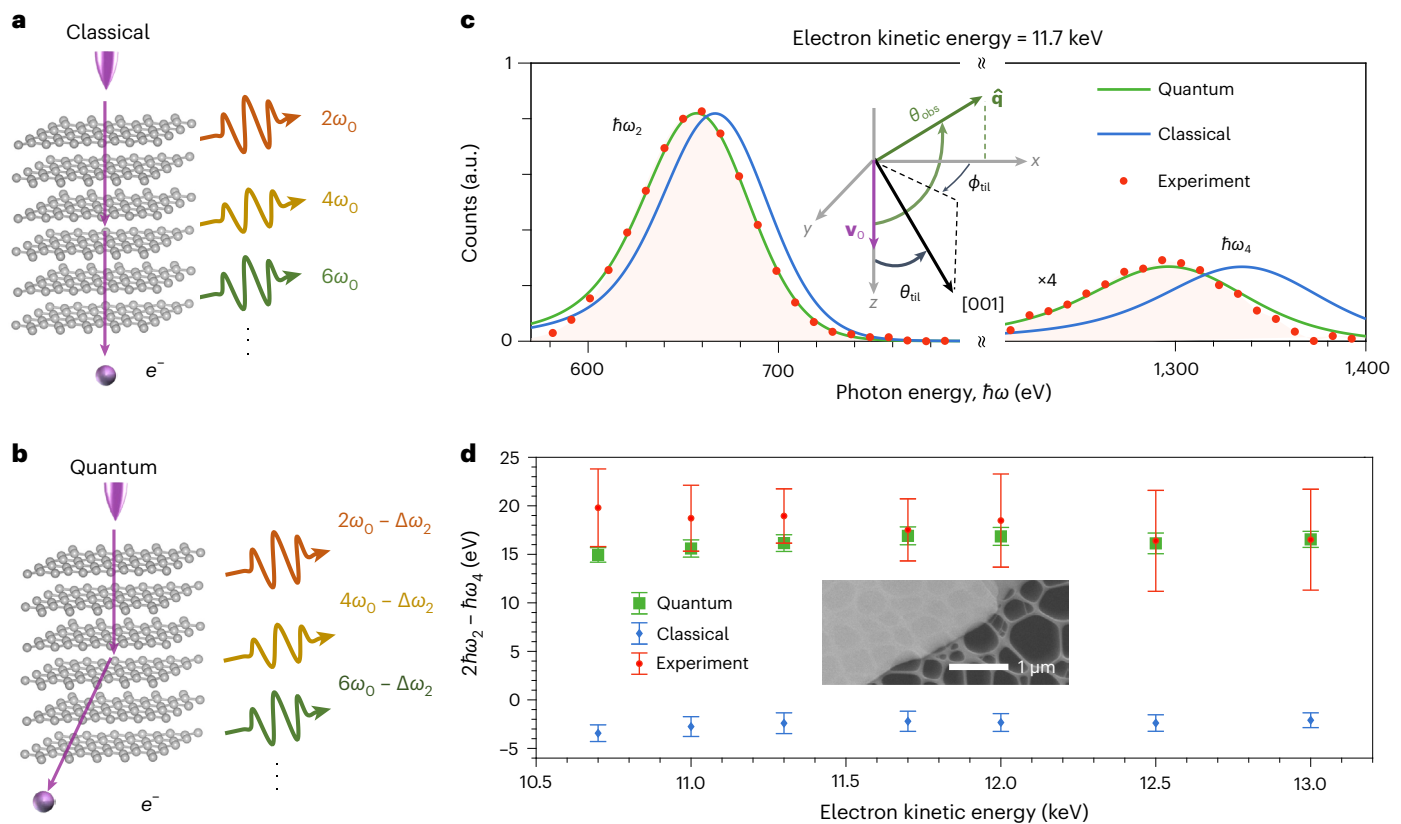


Fig. 1 | Quantum recoil and multimode X-ray generation in Smith–Purcell radiation. **a**, Classically, the electron is assumed to be undeflected for the purpose of determining the emitted photon frequency values. This results in uniformly spaced X-ray peaks where the output harmonics are given by $\omega_c = m\omega_0$, where ω_0 is the fundamental frequency. **b**, Energy and momentum conservation, however, results in a quantum shift $-\Delta\omega_m$ (which is negative in absolute sign) in the actual output photon energy, resulting in generally non-uniform X-ray peaks at $\omega = \omega_c - \Delta\omega_m$. **c**, Experimental measurements of the emitted X-ray spectrum (red symbols) are compared with the results of quantum recoil theory (green), showing excellent agreement. The left and right peaks (labelled ' $\hbar\omega_2$ ' and ' $\hbar\omega_4$ ', respectively) correspond to the (002) and (004) reciprocal lattice vectors of graphite, respectively. The blue curve is the result of classical theory, which neglects quantum recoil. The inset shows the coordinate system used in this work, with $\mathbf{v}_0$ denoting the initial velocity of the incident electron. **d**, Values of the difference, $2\hbar\omega_2 - \hbar\omega_4$, where experimental measurements (red symbols) are contrasted against the results of quantum (green) and classical (blue) theory for

different electron energies. In all theoretical predictions, electron-scattering events beyond Smith–Purcell radiation have been taken into account via Monte Carlo simulations. The experimental error bars reflect 95% confidence bounds in the Gaussian fitting of the X-ray peaks. The theoretical error bars reflect 95% confidence bounds in the Gaussian fitting of the Monte Carlo simulations, the convergence of the Monte Carlo simulations (relative error $<10^{-5}$) and the uncertainty of the sample tilt angles (see Methods for details). The experimental results closely favour the predictions of quantum theory over those of classical theory. Our results demonstrate that quantum recoil is measurable on a tabletop scale, at room temperature, and is robust against the presence of electron-scattering events. The inset shows an SEM image of the graphite sample (thickness 85 nm) on a TEM grid, used in the experiments presented here. The sample sizes used to determine the error bars for the peak photon energies experimentally and theoretically, respectively, are 121 and 1,002. Data are presented as mean values with error bars denoting the 95% confidence bounds.

An electron travelling through the crystalline van der Waals material will also experience elastic and inelastic scattering beyond Smith–Purcell radiation. To account for the impact of these scattering events, we perform Monte Carlo simulations to determine the change in velocity of each electron as it travels through the sample material. Between two adjacent scattering events, the Smith–Purcell radiation spectrum can be determined from equations (1), (3) and (5). The Smith–Purcell emission from an electron travelling a distance L between two adjacent scattering events has a bandwidth

$$\Delta\omega \approx \sqrt{\left(5.6 \frac{\beta c}{L}\right)^2 + \left(\frac{\partial\omega}{\partial\theta_{\text{obs}}} \Delta\theta_{\text{obs}}\right)^2} + (\Delta\omega_{\text{EDS}})^2, \quad (6)$$

where ω is the output photon angular frequency (set to ω_c for the classical case, that is, equation (1)). The first term (under the square root) in equation (6) corresponds to the intrinsic bandwidth, which is around 20 eV in our case. The second term accounts for the finite range of observation directions admitted by the angular aperture of

the X-ray detector. The final term accounts for the energy resolution of the X-ray detector, where ω_{EDS} is the X-ray frequency measured by the EDS detector and $\hbar\Delta\omega_{\text{EDS}} \approx 52$ eV (72 eV) is the photon energy resolution of the EDS detector for a photon energy of around 650 eV (1,300 eV). The detector energy resolution thus dominates the total bandwidth in our experiments. The total radiation spectrum is obtained by summing over all radiation spectra between any two adjacent scatterings contributed by all incident electrons. Broadening of the X-ray emission peaks due to electron scattering and electron-beam divergence are thus taken into account via the Monte Carlo simulations. More details on the Monte Carlo simulations are given in the Methods and Supplementary Section 3.

Figure 1c shows the multimode X-ray spectrum generated by an 11.7 keV electron beam impinging on a graphite single crystal. The filled red symbols are the experimental results. The blue and green lines denote the theoretical results corresponding to the classical (equation (1)) and quantum (equation (3)) models, respectively, at the observation angle $\theta_{\text{obs}} = 130^\circ$ and van der Waals material tilt angles

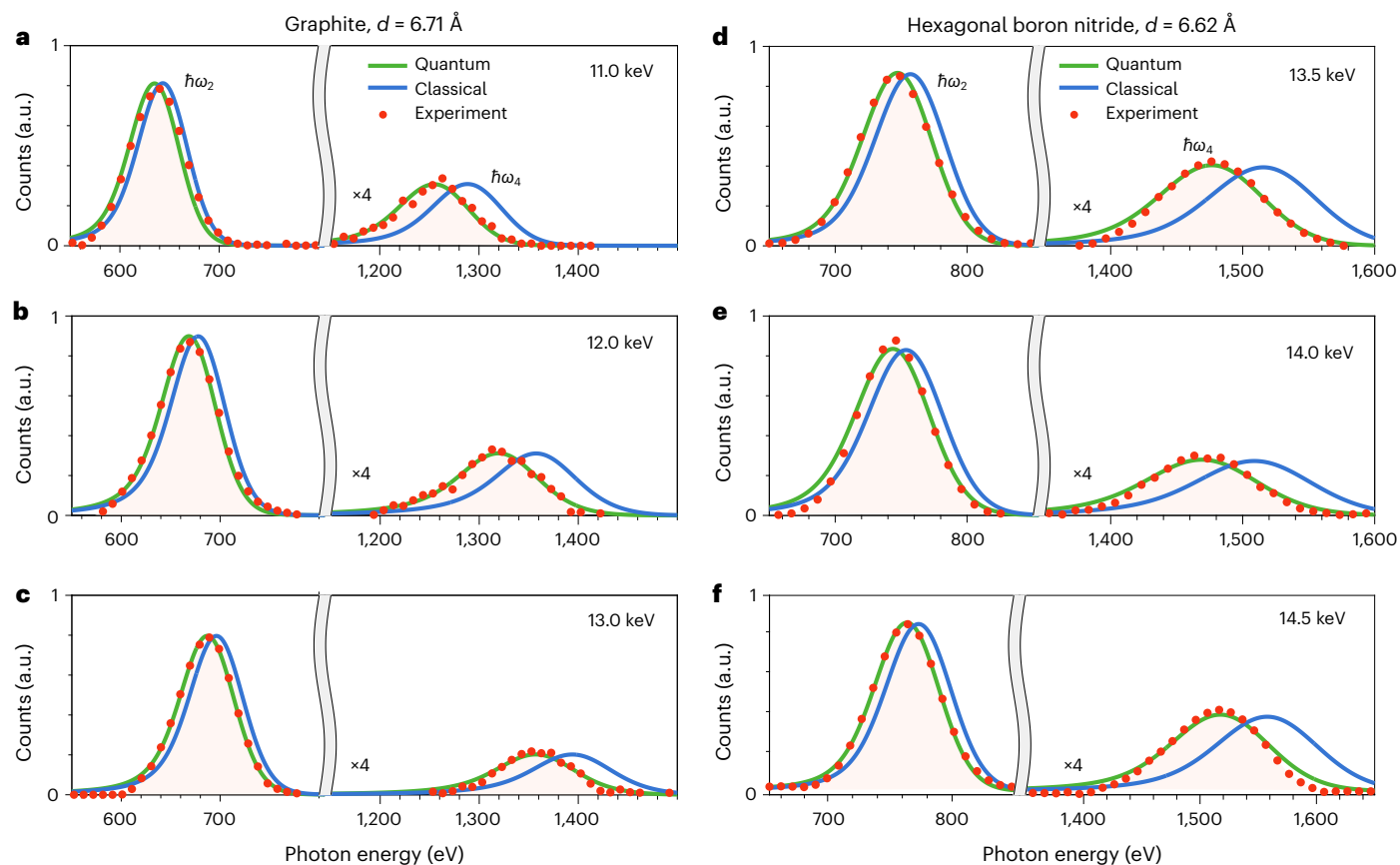


Fig. 2 | Quantum recoil for free electrons interacting with graphite and hexagonal boron nitride, respectively. **a–f**, The X-ray spectra containing the second- and fourth-order Smith–Purcell peaks for graphite (**a–c**) and hexagonal boron nitride (**d–f**) in each case for different incident electron kinetic energies

(as labelled). In each panel, the experimental measurements of the X-ray output (red) are compared with theoretical predictions in the presence (green) and absence (blue) of quantum recoil, showing excellent agreement between the experiment and quantum theory.

$\theta_{\text{til}} = 12.45^\circ$ and $\phi_{\text{til}} = 0.3^\circ$. The X-ray spectrum contains two distinct photon energy peaks, corresponding to the second-order ($m = 2$) and fourth-order ($m = 4$) Smith–Purcell photons. The experimental results are in excellent agreement with quantum theory, whereas they show noticeable discrepancies from the predictions of classical theory. The full X-ray spectrum, details of the multi-Gaussian peak deconvolution and background subtraction to yield the data presented in Fig. 1c are given in Supplementary Section 4. Details of the X-ray detector calibration are given in Supplementary Section 5.

As another way of analysing the photon energy shift induced by quantum recoil, we consider the energy difference between the second-order ($\hbar\omega_2$) and fourth-order ($\hbar\omega_4$) emission peaks. In the classical picture, equation (1) yields $2\hbar\omega_2 - \hbar\omega_4 = 0$, which is independent of all parameters including θ_{til} and the electron energy. Equation (3), which includes the effects of quantum recoil, shows that the difference generally takes on a non-zero value that depends on the choice of experimental parameters. The experimental results shown in Fig. 1d are indicated by the filled red circles, with error bars determined by Gaussian fitting of the measured Smith–Purcell peaks. The good agreement between the experimental results and the quantum theoretical predictions confirms that quantum recoil is indeed measurable using a combination of tabletop electron sources and crystalline materials. In this case, electron energies between 10 and 15 keV were used. Both the classical and quantum predictions have taken electron-scattering events beyond Smith–Purcell radiation into consideration via Monte Carlo simulations. Similar results for 55-nm-thick graphite and 95-nm-thick hexagonal boron nitride are shown in Supplementary Section 6, giving further evidence that our

quantum recoil measurements and conclusions are robust to the choice of material and sample thickness. The good agreement between experimental results and quantum theory also affirms that quantum recoil is robust against the presence of other scattering events, even in thick samples. Our results constitute a measurement of quantum recoil in free-electron spontaneous emission. Importantly, they also constitute the observation of a quantum phenomenon at room temperature, and while requiring only a tabletop source of free electrons.

In our experiments, hexagonal boron nitride and graphite thin films are mechanically exfoliated from their bulk counterparts and transferred to Au grids using the wet-transfer method. For these thin films, the surface layer belongs to the (001) family of planes. The tilt angles θ_{til} and ϕ_{til} , that is, the polar and azimuthal angles of the [001] zone axis, respectively, are usually not zero since the Au grid is not atomically flat. Furthermore, some areas of the sample are bent during the sample preparation and/or during the loading of the Au grid on to the sample holder of the SEM. The sample tilt angles can be accurately determined with a transmission electron microscope (TEM) using Kikuchi lines²², where the value of θ_{til} is within 10° and the value of ϕ_{til} can be any value from 0 to 2π . However, this only gives us an estimate of the actual sample tilt angles in the SEM. To accurately determine the sample tilt angles in the SEM itself, we perform a fitting step by matching the measured peak photon energies with the theoretical predictions. To ensure this does not unfairly favour our hypothesis that the experiment should match quantum theory we use the classical equations to fit the experimental data for the classical theory predictions, and we use the quantum equations to fit the experimental data for the quantum theory predictions. Importantly, we find that only the

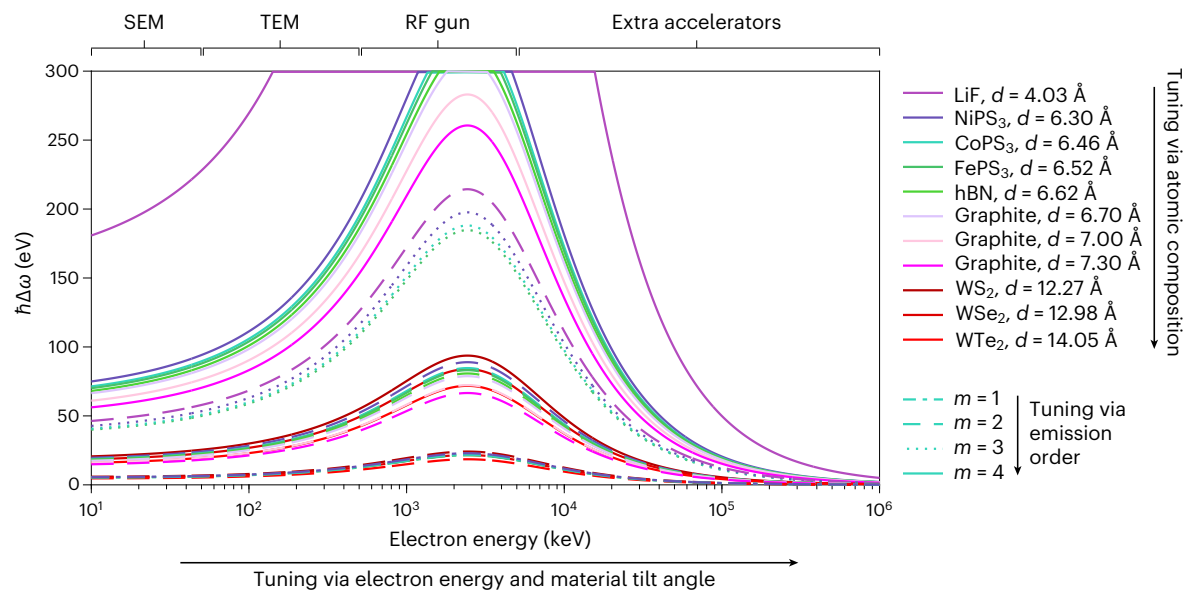


Fig. 3 | Tuning quantum recoil in Smith–Purcell radiation. For a given classical emitted photon energy (300 eV in this case, observed at $\theta_{\text{obs}} = 10^\circ$, $\phi_{\text{obs}} = 0^\circ$), the quantum recoil—which directly determines the actual output photon energy—can be tuned by varying the electron energy in combination with the tilt angle, by varying the atomic composition of the crystalline material (the different coloured lines) or by measuring the emitted photons at different radiation orders (the different line styles). The quantum recoil $\hbar\Delta\omega$ is defined as the difference between the classical photon energy and the actual (quantum) photon energy. Exemplary instrumentation corresponding to various ranges of electron

energies is listed along the top, that is, SEM, TEM and radiofrequency (RF) gun (still laboratory scale). Larger electron energies require additional accelerator stages that result in much larger setups, along with the need for bulky neutron shielding at electron energies of >10 MeV. We see that with existing materials and tabletop electron energies, it is possible to achieve actual emitted photon energies from approximately the classical prediction of 300 eV all the way down to extremely low photon energies by adjusting the amount of quantum recoil. Here $\phi_{\text{til}} = 0^\circ$ and the value of θ_{til} for each point is given in Supplementary Fig. 13. hBN, hexagonal boron nitride.

results of quantum theory agree with the experimental measurements (for example, Fig. 1d), with the least-squares error of the fitting being over ten times superior for the quantum case than for the classical case. The uncertainty associated with the fittings is also included in the error bars of Fig. 1d. More details are described in the Methods and Supplementary Section 7.

The X-ray spectra for graphite (compared against classical and quantum predictions) that correspond to various points in Fig. 1d are shown in Fig. 2a–c. Figure 2d–f shows the results obtained from hexagonal boron nitride (the figure corresponding to Fig. 1d for this set of measurements is presented in Supplementary Section 6). In all cases, we see that the experimental measurements confirm the importance of taking quantum recoil into account when predicting the photon energy output from Smith–Purcell radiation.

To give an idea of the X-ray intensity generated by our experiments, the measured X-ray intensity is about 0.012 pW for the second-order X-rays and 0.004 pW for the fourth-order X-rays in Fig. 2f. These measurements were performed using an average of current of about 4.4 nA at an observation angle of $119.0 \pm 5.5^\circ$. The corresponding photon flux is 3.4×10^3 photons $\text{s}^{-1} \text{sr}^{-1}$ for second-order X-rays and 5.0×10^2 photons $\text{s}^{-1} \text{sr}^{-1}$ for fourth-order X-rays. In units of photons $\text{electron}^{-1} \text{sr}^{-1}$, this corresponds to 1.2×10^{-7} photons $\text{electron}^{-1} \text{sr}^{-1}$ for the second-order X-rays and 1.8×10^{-8} photons $\text{electron}^{-1} \text{sr}^{-1}$ for the fourth-order X-rays. Note that these estimates of intensity and photon flux only include the photons detected by the employed setup and not the photons emitted at all angles. Although the main purpose of our study is not to demonstrate a commercially ready source of X-rays, we note that by scaling up the current up to 5 mA, which is typical in conventional X-ray tubes, we achieve an X-ray flux of about 3.9×10^9 photons $\text{s}^{-1} \text{sr}^{-1}$ for second-order X-rays and about 5.7×10^8 photons $\text{s}^{-1} \text{sr}^{-1}$ for fourth-order X-rays. If we take the distance between the X-ray source and an X-ray imaging detector (for example, a charge-coupled device) to be 10 cm, the corresponding X-ray intensity

at the detector is 3.9×10^5 photons $\text{s}^{-1} \text{mm}^{-2}$ for second-order X-rays and about 5.7×10^4 photons $\text{s}^{-1} \text{mm}^{-2}$ for fourth-order X-rays. Such X-ray intensities have been shown to be adequate for imaging applications using sufficiently thin samples⁴³.

Due to energy conservation, quantum recoil not only induces energy shifts in the emitted photons, but also affects the quantized energy loss in the outgoing electrons. The emitted photon energy and the corresponding electron energy loss are entangled via energy conservation, potentially also allowing us to observe quantum recoil by measuring the post-selected electron energy loss spectrum (see Supplementary Section 8 for details). In particular, measuring the post-selected electron energy loss spectrum would involve recording the outgoing electron energies if and only if outgoing photons corresponding to the shifted photon energy peaks are simultaneously detected. This is more cumbersome than (but equivalent to) directly measuring the outgoing photon energy shifts, as we have done in this work. It should also be noted that this post-selected electron energy loss spectrum would feature peaks that are shifted as compared to the peaks predicted by classical theory. Therefore, simply measuring the quantized energy loss of the electrons does not constitute an observation of quantum recoil, unless the presence of quantum recoil-induced energy shifts can be verified. Our success in measuring quantum recoil in Smith–Purcell radiation motivates the investigation of similar quantum mechanical effects in inverse processes where optical waveforms are used to control or accelerate electrons^{18,23,37}.

Figure 3 illustrates the tuning of quantum recoil via various degrees of freedom. We see that a wide range of quantum recoil tuning can be achieved, even by confining ourselves to tabletop electron energies: from practically zero quantum recoil, to quantum recoil so strong that a classically predicted soft X-ray photon emerges as an extremely low-energy photon. The tuning can be performed by varying the electron kinetic energy and material tilt angle, by changing the atomic composition of the target material, or by measuring the

radiation emission of different orders m , where m refers to radiation corresponding to the $(00m)$ reciprocal lattice vector. Although Fig. 3 considers the specific case of a classically predicted 300 eV photon, the same principles and conclusions will apply to photons of other energies (see Supplementary Section 9 for examples of other classically predicted photon energies). Here, multiple values of the interlayer spacing d are used for graphite to reflect the fact that its interlayer spacing can be tuned by controlling the conditions of its synthesis⁴⁴, giving us flexibility even with one type of material. Importantly, our findings indicate that conventional tabletop electron sources with kinetic energies of 1–300 keV (which include SEM and TEM energies), in combination with crystalline materials, are promising platforms for investigating quantum recoil in free-electron spontaneous emission.

Discussion

We have shown that quantum recoil in free-electron spontaneous emission can be measured and tuned using a laboratory-scale platform. This platform is useful for exploring new phenomena as well as realizing intriguing science in the regime of substantial quantum recoil. Our findings directly corroborate the effects arising in light–matter interactions due to quantization of the electromagnetic field. These effects were first analysed in the context of Cherenkov radiation by Ginzburg²⁵ and Sokolov⁴⁰, and have been predicted to cause quantum recoil in free-electron spontaneous emission^{39,42}. Our findings provide experimental evidence of quantum recoil in the context of the Smith–Purcell effect. This also shows that a tabletop platform can already be used to make important headway into these studies. Together with higher-order quantum electrodynamics processes⁴⁵ and the quantum-wave nature of the electron (for example, arising from shaping of the electron wavefunction spatially and temporally)^{42,46–48}, measuring quantum recoil paves the way to studying the fundamental physics and potential technological advances that arise from quantum aspects of electron–photon interactions⁴⁹.

Our findings reveal that quantum recoil is substantial and must be given due consideration even in the non-relativistic limit, especially when nanoscale and subnanoscale periodicities are involved. This is important, especially with increasing interest in nanoscale free-electron-driven radiation sources, along with the use of low-energy electrons for chip-scale integration^{23,50,51}. Our theory and experiments focus on the spontaneous emission that results from free-electron interactions with the nanometre and subnanometre gratings provided by the periodic lattices of crystalline solids. This is complementary to existing analyses on free-electron spontaneous emission using grating periods of the order of 10 nm and larger where the quantum-recoil-induced photon energy shift is theoretically investigated³⁹. Future work will include the study of quantum recoil in the context of related light–matter interactions, such as the inverse Smith–Purcell effect and the inverse Cherenkov effect.

It should be noted that any type of crystalline material—not just van der Waals materials—could have served as the target material in our platform. However, the advantage of using van der Waals materials arises from the large variety of compound combinations to enable precise control over the lattice constants that affect the quantum recoil⁵². To eliminate most electron-scattering events in experiments, one can use a target material with a thickness less than the electron mean free path (about 10 nm for graphite subjected to an electron energy of ~10 keV). Such a small thickness, however, corresponds to a short interaction length that could make measuring the Smith–Purcell signal challenging. This could be addressed by increasing the kinetic energy of the electron to increase the mean free path and hence the interaction length.

In summary, our theory and experimental measurements show that quantum recoil in Smith–Purcell radiation can be substantial, even when driven using conventional tabletop electron sources like SEMs and TEMs. We show that tabletop electron sources can thus serve as

robust, accessible platforms for investigating the quantum aspects of electron–photon interactions. Our experimental findings reveal that quantum recoil is measurable at room temperature and is robust against the presence of electron-scattering events beyond Smith–Purcell radiation, making it a promising means for the precise tailoring of output photon spectra or post-selected output electron spectra. Potential applications also include the study of radiation reaction, where crystalline solids have been used for measurements in the regime of ultrarelativistic electron energies⁵³ and X-ray quantum optics⁴⁹.

Online content

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Methods

Monte Carlo simulations

Monte Carlo simulations to capture the impact of electron-scattering events beyond Smith–Purcell radiation on our measurements were carried out using the CASINO software (details in Supplementary Section 3). Unless otherwise specified, these simulations were performed for both classical theory and quantum theory predictions. The error bars in plots like Fig. 1d capture the 95% confidence bounds in the Gaussian fitting of the Monte Carlo simulations, the convergence of the Monte Carlo simulations (relative error $<10^{-5}$) and the uncertainty of the sample tilt angle. The convergence of the Monte Carlo simulations was determined by increasing the number of particles used (by 10,000 each time) until the relative change in the output (which is the peak photon energy in our case) falls below 10^{-5} . On average, we found that 50,000 particles were needed to achieve convergence in this way. The uncertainty of the sample tilt angle arises from the practical impossibility of exactly determining the tilt angle of the sample using our SEM apparatus. However, we are able to get around this by fitting (using a method of least squares) for the tilt angle, which we do separately with classical theory and with quantum theory, not only to obtain the precise value of the tilt angle but more importantly to see which theory can provide a superior fit to the experimental results. As we see in Supplementary Section 7, a much better fit is obtained using quantum theory compared with classical theory. This conclusion continues to be borne out when we apply the best possible fitted angles in classical and quantum scenarios, respectively, to the classical and quantum theories in Fig. 1d: once again, the experimental results confirm that quantum theory is accurate and that we have broken into a regime where quantum recoil is measurable.

Sample preparation

Single-crystal hexagonal boron nitride was synthesized using the atmospheric pressure metal flux method⁵⁴. Thin films of hexagonal boron nitride and graphite are exfoliated mechanically on to silicon substrates (a 285 nm SiO₂ film) and transferred to Au grids using the wet-transfer method. The Au grid is held in a TEM grid holder for SEM during the X-ray measurements in SEM. The benefit of using a TEM grid instead of a regular substrate is that the TEM grid minimizes background radiation from free electrons travelling through the sample. The thickness of the graphite and hexagonal boron nitride samples used in Figs. 1 and 2 is 85 ± 2 nm and 95 ± 2 nm, respectively, which is measured via convergent beam electron diffraction (see Supplementary Section 10 for details).

X-ray measurements

The quantum recoil measurements (Fig. 1 and Fig. 2a–c) were performed using a SEM (EPMA) JEOL JXA-8530F instrument, and the measurements for Fig. 2d–f were performed using a JEOL JSM-7800F Prime field-emission SEM. The emitted X-ray spectra were measured using a silicon drift energy dispersive X-ray spectroscopy detector in the SEM. The X-ray detector was calibrated to measure the X-ray photon energies with an accuracy within ± 0.5 eV by measuring the K α peaks of C, N, O, F, Mg, Al and Si (see Supplementary Section 5). The take-off angle θ_{obs} and the observation angle range $\Delta\theta_{\text{obs}}$ are 130° (119°) and 1.67° (11°), respectively, for the JEOL JXA-8530F (JSM-7800F Prime). The energy resolution of the X-ray detector on the JEOL JXA-8530F instrument is $\hbar\Delta\omega_{\text{EDS}} \approx 52$ eV (72 eV) for a photon energy of around 650 eV (1,300 eV), and $\hbar\Delta\omega_{\text{EDS}} \approx 50$ eV (65 eV) for a photon energy of around 650 eV (1,300 eV) for the JSM-7800F Prime instrument. The energy resolution is estimated via measuring the bandwidth of characteristic peaks and interpolating the data in our X-ray photon energy range of interest. The effect of refraction on the X-ray photon exiting the van der Waals crystal is taken into account, as discussed in Supplementary Section 11. Each experimental X-ray spectrum in Figs. 1c and 2 was the result of an approximately four-hour measurement. The background radiation of the measured spectra was subtracted using

the NIST DTSA-II software^{55,56}. The overlapped peaks are deconvoluted into separated peaks via multi-Gaussian function fitting, with the 95% confidence bounds of this fitting being reflected in the error bars of plots like Fig. 1d (see Supplementary Section 4 for details).

Data availability

The data represented in Figs. 1–3 are available as Supplementary Information files. All other data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request. Source data are provided with this paper.

Code availability

All code that support the plots within this paper are available from the corresponding authors upon reasonable request.

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Author contributions

S.H. led the project, designed and performed the electron microscopy experiments and analysed the data. R.D. and Z.L. prepared the samples. N.P. and S.H. developed the theory and performed the simulations. J.S.H. and C.B. contributed to the analysis of the experimental results. S.H. and L.J.W. wrote the paper, with inputs from all other authors. L.J.W. conceived the idea and supervised the project.

Competing interests

The authors declare no competing interests.

Additional information

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